The sharp index for sums of arithmetic and geometric means

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Task 1: the complete maximal-index problem

For a probability vector $p = (p_1, \dots, p_d)$, form the two lists of d^2 entries

$$a_{ij} = \frac{p_i + p_j}{2}, \qquad g_{ij} = \sqrt{p_i p_j}.$$

Let $A_h(p)$ and $G_h(p)$ denote the sums of the h largest entries in the respective lists, each sorted independently. Set $A_0 = G_0 = 0$, and define

$$H(d) = \max\{h \in \{0, \dots, d^2\} : A_h(p) + G_h(p) \le d \text{ for every probability vector } p\}.$$

Caro and Graswald conjectured the value $2d - 5$ in Conjecture IV.19 of their published paper [1], numbered 4.19 in the current preprint. We prove that value throughout its intended large-dimension range and determine all small cases.

Theorem 1. *The exact values are*

$$\boxed{H(1) = 0, \quad H(2) = 1, \quad H(3) = 2, \quad H(4) = 4, \qquad H(d) = 2d - 5 \quad (d \ge 5).}$$

For $d \ge 5$, the inequality at $h = 2d - 5$ is strict for every p. For every such d, an explicit rational probability vector violates the inequality at $h = 2d - 4$.

The unrestricted printed formula needs the small-dimension qualifications above. The substance of the result is the uniform proof and sharpness for all $d \ge 5$, beyond that correction.

1. Two elementary tools

For every h, A_h is convex in p: it is the maximum over all h-element sets $S \subseteq \{1, \dots, d\}^2$ of the linear functions $\sum_{(i,j)\in S}(p_i + p_j)/2$. It is also invariant under permuting the coordinates of p.

Moreover,

$$\sum_{i,j} g_{ij}^2 = \sum_{i,j} p_i p_j = 1, \qquad G_h \le \sqrt{h}. \tag{1}$$

If $x = \max_i p_i$, then the largest entry of g is x. Applying Cauchy–Schwarz to the other $h - 1$ selected entries gives

$$G_h \le x + \sqrt{(h-1)(1-x^2)} \qquad (h \ge 1). \tag{2}$$

These arguments make no assumption that the two entry lists have the same ordering.

2. Sufficiency in every dimension $d \geq 5$

Fix $h = 2d - 5$ and order $p_1 \geq \cdots \geq p_d$. Put $x = p_1$.

Case $x \leq 2/3$. Consider the polytope

$$P = \{p \in [0, 2/3]^d : \ \textstyle\sum_i p_i = 1\}.$$

At a vertex of P, at most one coordinate lies strictly between 0 and $2/3$. Indeed, if two coordinates did, a sufficiently small transfer between them in either direction would express the point as the midpoint of two distinct points of P. The sum constraint therefore forces every vertex to be a permutation of $v = (2/3, 1/3, 0, \ldots, 0)$. Every point of this bounded polytope is a convex combination of its vertices, so convexity and symmetry give $A_h(p) \leq A_h(v)$.

The largest arithmetic entries at v are $2/3, 1/2, 1/2$, followed by $1/3$ repeated $2d - 3$ times. In particular,

$$A_h(p) \leq \frac{2}{3} + 2\frac{1}{2} + \frac{h-3}{3} = \frac{h+2}{3} = \frac{2d-3}{3}.$$

Combining this with (1),

$$A_h(p) + G_h(p) \leq \frac{2d-3}{3} + \sqrt{2d-5} < d. \tag{3}$$

For the strict inequality, both sides of $\sqrt{2d-5} < (d+3)/3$ are nonnegative and

$$(d+3)^2 - 9(2d-5) = (d-6)^2 + 18 > 0.$$

Case $x \geq 2/3$. Every arithmetic entry outside the first row and column is at most $1 - x$, whereas every entry in that row or column is at least $x/2$. Since $1 - x \leq x/2$, the top h arithmetic entries may be chosen within that row and column, including ties. Because $h = 1 + 2(d-3)$, they consist of $a_{11} = x$ and the $d - 3$ largest symmetric pairs. Hence

$$A_h(p) = (d-2)x + \sum_{j=2}^{d-2} p_j \leq 1 + (d-3)x.$$

By (2) and Cauchy–Schwarz in $\mathbb{R}^2$,

$$\begin{aligned} A_h(p) + G_h(p) &\leq 1 + (d-2)x + \sqrt{2(d-3)(1-x^2)} \\ &\leq 1 + \sqrt{(d-2)^2 + 2(d-3)} \\ &= 1 + \sqrt{d^2 - 2d - 2} < d. \end{aligned} \tag{4}$$

The last step follows from $(d-1)^2 - (d^2 - 2d - 2) = 3 > 0$. Together, the cases prove $H(d) \geq 2d - 5$, including strictness.

3. Sharpness in every dimension $d \geq 5$

Write $k = d - 3 \geq 2$ and $D = (k+2)^2 + k$. Take

$$p = \frac{1}{D}\big((k+2)^2, \underbrace{1, \ldots, 1}_{k \text{ entries}}, 0, 0\big), \qquad x = \frac{(k+2)^2}{D}. \tag{5}$$

At $h = 2d - 4 = 2k + 2$, select for the arithmetic sum the entry $(1, 1)$, the $2k$ entries $(1, j), (j, 1)$ with $2 \le j \le k + 1$, and one entry joining coordinate 1 to a zero coordinate. Their sum is

$$\left(k + \frac{3}{2}\right) x + \frac{k}{D}.$$

For the geometric sum, select $(1, 1)$ and the same $2k$ entries, and then $(2, 2)$. Their sum is

$$x + \frac{2k(k + 2)}{D} + \frac{1}{D}.$$

The largest-entry sums dominate these selected sums, so

$$A_{2d-4}(p) + G_{2d-4}(p) - d \ge \frac{k^2 - 2}{2D} > 0. \tag{6}$$

This calculation does not require identifying either full ordering. Since all entries are nonnegative, failure at this index implies failure at every larger index. Thus $H(d) = 2d - 5$ for every $d \ge 5$.

4. The four small dimensions

For $d = 1$, the only vector is (1) and the sum at $h = 1$ is $2 > 1$, so $H(1) = 0$. For $d = 2$, the sum at $h = 1$ is $2 \max_i p_i \le 2$; at $h = 2$, $p = (1, 0)$ gives $A_2 + G_2 = 5/2 > 2$.

For $d = 3$, convexity on the full probability simplex gives $A_2(p) \le A_2(1, 0, 0) = 3/2$. Therefore $A_2 + G_2 \le 3/2 + \sqrt{2} < 3$. At $h = 3$, the vector $(4/5, 1/5, 0)$ has $A_3 = 9/5$ and $G_3 = 8/5$, whose sum exceeds 3. This proves $H(2) = 1$ and $H(3) = 2$.

For $d = 4$, order the coordinates decreasingly. Directly identifying the first four entries in each list yields

$$A_4 + G_4 = 3p_1 + p_2 + 2\sqrt{p_1 p_2} + \max\left(p_2, \frac{p_1 + p_3}{2}\right) + \max\left(p_2, \sqrt{p_1 p_3}\right).$$

Indeed, the first three entries in either list consist of $(1, 1)$ and the pair $(1, 2), (2, 1)$; the next entry is the larger of $(2, 2)$ and $(1, 3)$. Ties cause no change in the sum. With $u = (\sqrt{p_1}, \sqrt{p_2}, \sqrt{p_3})^T$, this expression is the maximum of $u^T M_i u$, for the following four matrices:

$$M_1 = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 3 & 1 & 1/2 \\ 1 & 2 & 0 \\ 1/2 & 0 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 7/2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}, \quad M_4 = \begin{pmatrix} 7/2 & 1 & 1/2 \\ 1 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

All four satisfy $M_i \le 4I$ in the positive-semidefinite order. For M_1 and M_3, the complementary quadratic forms are $(u_1 - u_2)^2 + 4u_3^2$ and $\frac{1}{2}(u_1 - 2u_2)^2 + \frac{7}{2}u_3^2$, respectively. For $4I - M_2$ the leading principal minors are $1, 1, 7/2$; for $4I - M_4$ they are $1/2, 1/2, 1$, all positive. Consequently $A_4 + G_4 \le 4\|u\|^2 \le 4$.

Finally, at $h = 5$, the vector $(3/4, 1/12, 1/12, 1/12)$ gives

$$A_5 = \frac{29}{12}, \qquad G_5 = \frac{7}{4}, \qquad A_5 + G_5 = \frac{25}{6} > 4.$$

This proves $H(4) = 4$ and completes Theorem 1. The index-four bound is attained, for example, at $(1/2, 1/2, 0, 0)$ and $(4/5, 1/5, 0, 0)$.

Verification and scope

The accompanying `verify_index.py` checks five general symbolic identities, the four quadratic-form identities and all 28 principal minors, the four small counterexamples, two index-four equality cases, and the independently sorted rational counterexamples in dimensions 5 through 60. It also checks 7,997 vectors of the form $p_i = r_i^2 / \sum_j r_j^2$, with sorted integer $r_i \in \{0, \ldots, 5\}$, in dimensions 1 through 10. These finite checks are diagnostics; the proof for all dimensions and all real probability vectors is the argument above. Convexity, polytope vertices, Cauchy–Schwarz, and the positive-semidefinite criterion are classical tools. This is an authoring-agent proof audit, not independent peer review.

Tasks 2–5: proposed follow-up questions

These are unproved directions, proposed after the full solution of Task 1; they earn no additional solution credit.

2. Determine the exact value of $\sup_p[A_{2d-5}(p) + G_{2d-5}(p)]$ for each $d \geq 5$, and classify all maximizing probability vectors.

3. Determine the sharp asymptotic size of $d - \sup_p[A_{2d-5}(p) + G_{2d-5}(p)]$ as $d \to \infty$, and establish stability estimates for vectors near that supremum.

4. Determine the optimal index in the original Ky Fan norm inequality of Remark IV.18 of [1], exploiting the full operator expression before the triangle-inequality reduction.

5. For each $\lambda > 0$, determine the largest h such that $A_h(p) + \lambda G_h(p) \leq d$ for every probability vector, including the transition values of λ.

References

[1] M. C. Caro and B. R. Graswald, *Necessary criteria for Markovian divisibility of linear maps*, J. Math. Phys. **62**, 042203 (2021), doi:10.1063/5.0031760. Current preprint: arXiv:2009.06666v3 (30 October 2022), Conjecture 4.19.